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Experiment 1 – Design Star and Snowflake Schema

Aim

To design Star Schema and Snowflake Schema for an Online Music Streaming Company using suitable fact tables and dimension tables.

Objective

- To understand the concept of Data Warehouse Schema.
- To design a Star Schema.
- To design a Snowflake Schema.
- To understand the relationship between Fact Tables and Dimension Tables.
- To analyze music streaming data efficiently.

Theory

What is a Data Warehouse Schema?

A Data Warehouse Schema is the logical blueprint that shows how data is organized, connected, and used for analysis in a data warehouse. It defines tables, keys, relationships, and rules that help Business Intelligence (BI) tools perform analytical queries efficiently.

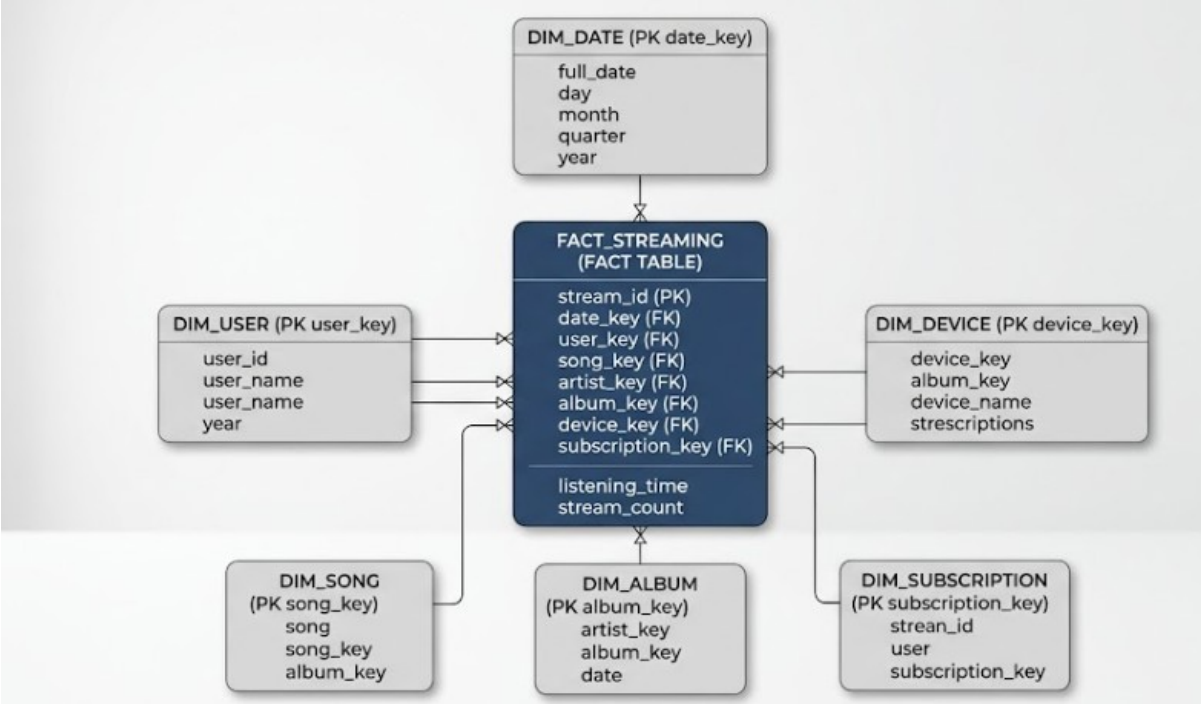
It includes:

- Fact Tables: Store measurable business events such as streams, subscriptions, or revenue.
- Dimension Tables: Store descriptive information such as user, artist, song, album, and date.
- Relationships: Connect fact tables with dimension tables using foreign keys for efficient querying.

1. Star Schema

A Star Schema organizes data around a central Fact Table connected directly to multiple Dimension Tables. It is simple, fast for querying, and commonly used in data warehouses.

Structure of Star Schema (Music Streaming Activity)



Fact Table

Streaming_Fact

Column	Description
stream_id	Primary Key
date_key	Foreign Key
user_key	Foreign Key
song_key	Foreign Key
artist_key	Foreign Key
album_key	Foreign Key
device_key	Foreign Key
subscription_key	Foreign Key
listening_time	Total listening duration
stream_count	Number of streams

Dimension Tables

Date_Dimension

Column	Description
date_key	Primary Key
full_date	Date
day	Day
month	Month
quarter	Quarter
year	Year
weekday	Weekday

User_Dimension

Column	Description
user_key	Primary Key

user_name	User Name
gender	Gender
age_group	Age Group
city	City
country	Country
subscription_type	Free/Premium

Song_Dimension

Column	Description
song_key	Primary Key
song_title	Song Name
genre	Genre
language	Language
duration	Song Duration
release_year	Release Year

Artist_Dimension

Column	Description
artist_key	Primary Key
artist_name	Artist Name
country	Country
debut_year	Debut Year
record_label	Record Label

Album_Dimension

Column	Description
album_key	Primary Key
album_name	Album Name
release_date	Release Date
album_type	Single/EP/Album

Device_Dimension

Column	Description
device_key	Primary Key
device_name	Mobile/Desktop
operating_system	Android/iOS/Windows
app_version	Application Version

Subscription_Dimension

Column	Description
subscription_key	Primary Key
plan_name	Free/Premium/Family
monthly_fee	Subscription Cost
payment_method	Card/UPI/Wallet

Structure of Star Schema (Revenue Analysis)

Fact Table

Revenue_Fact

Column	Description
revenue_id	Primary Key
date_key	Foreign Key

user_key	Foreign Key
subscription_key	Foreign Key
payment_key	Foreign Key
country_key	Foreign Key
revenue_amount	Revenue Generated

Dimension Tables

Payment_Dimension

Column	Description
payment_key	Primary Key
payment_mode	UPI/Card/Wallet
bank_name	Bank
transaction_status	Success/Failed

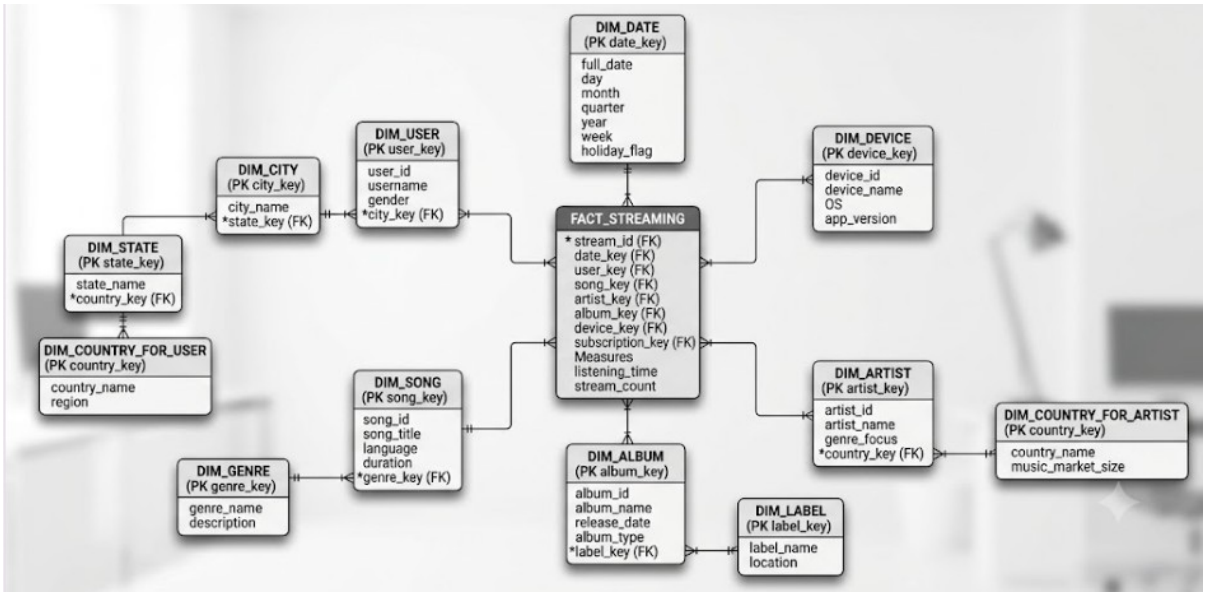
Country_Dimension

Column	Description
country_key	Primary Key
country	Country Name
region	Region
currency	Currency

2. Snowflake Schema

A Snowflake Schema is an extension of the Star Schema where dimension tables are normalized into smaller related tables. It reduces redundancy and saves storage space.

Structure



Streaming_Fact

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Song_Dimension — song_key, song_title, genre_key, album_key, artist_key

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Genre_Dimension — genre_key, genre_name

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Album_Dimension — album_key, album_name, label_key

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Record_Label_Dimension — label_key, label_name

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Artist_Dimension — artist_key, artist_name, country_key

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Country_Dimension — country_key, country, region

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User_Dimension — user_key, city_key, subscription_key

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City_Dimension — city_key, city_name, state_key

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State_Dimension — state_key, state_name, country_key

Main Difference

Star Schema	Snowflake Schema
• Simple design. • Dimension tables are denormalized. • Fewer joins. • Faster query execution. • Higher storage due to redundant data.	• More normalized structure. • Less redundancy. • Requires more joins. • Better storage utilization. • Slightly slower query performance.

Result

Thus, the Star Schema and Snowflake Schema for the Online Music Streaming Company were successfully designed by identifying the Streaming Activity Fact Table and Revenue Fact Table along with their corresponding Dimension Tables, enabling efficient data analysis and reporting.